

LLaMA：开放高效的基础语言模型

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Abstract

我们提出了 LLaMA，这是一系列参数规模从 7B 到 65B 的基础语言模型。我们在数万亿词元上训练模型，并证明了仅使用公开可用的数据集就能训练出最先进的模型，而无需依赖专有和不可访问的数据集。特别是，LLaMA-13B 在大多数基准测试上优于 GPT-3 (175B)，而 LLaMA-65B 与最好的模型 Chinchilla-70B 和 PaLM-540B 相比具有竞争力。我们将所有模型向研究社区发布¹。

1 引言

在海量文本语料上训练的大语言模型 (LLMs) 已经展示了其从文本指令或少量示例中执行新任务的能力 (?)。这些少样本特性首次出现在将模型扩展到足够规模时 (?)，从而催生了一系列关注进一步扩展这些模型的研究工作 (??)。这些努力基于一个假设：更多的参数将带来更好的性能。然而，最近的研究表明，对于给定的计算预算，最佳性能并非来自最大的模型，而是来自在更多数据上训练的较小模型。

所提出的缩放定律的目标是确定如何针对特定的计算预算最优地缩放数据集和模型规模。然而，这个目标忽略了推理预算，而当大规模部署语言模型时，推理预算变得至关重要。在此背景下，给定一个目标性能水平，首选的模型不是训练最快的，而是推理时最快的——尽管训练一个大模型来达到某一性能水平可能更便宜，但经过更长时间训练的较小模型最终在推理时会更经济。例如，尽管建议在 200B 词元上训练一个 10B 模型，但我们发现一个 7B 模型的性能在超过 1T 词元后仍然持续提升。

本研究的工作重点是训练一系列语言模型，通过使用比通常更多的词元进行训练，在各种推理预算下实现最佳性能。由此产生的模型被称为 LLaMA，参数规模从 7B 到 65B 不等，其性能与现有最好的 LLMs 相比具有竞争力。例如，LLaMA-13B 在大多数基准测试上优于 GPT-3，

尽管其规模小了 10 $\times$ 。我们相信该模型将有助于普及 LLMs 的使用和研究，因为它可以在单个 GPU 上运行。在规模的高端，我们的 65B 参数模型也与最好的大语言模型（如 Chinchilla 或 PaLM-540B）不相上下。

与 Chinchilla、PaLM 或 GPT-3 不同，我们仅使用公开可用的数据，这使得我们的工作与开源兼容，而大多数现有模型依赖的数据要么不公开，要么没有文档记录（例如“Books – 2TB”或“Social media conversations”）。存在一些例外情况，特别是 OPT (?)、GPT-NeoX (?)、BLOOM (?) 和 GLM (?)，但这些模型没有能与 PaLM-62B 或 Chinchilla 相竞争的。

在本文的其余部分，我们将概述我们对 transformer 架构 (?) 所做的修改，以及我们的训练方法。然后，我们将报告模型的性能，并在一系列标准基准测试上与其他 LLMs 进行比较。最后，我们利用负责任 AI 社区提供的一些最新基准测试，展示我们模型中存在的一些偏见和毒性。

2 方法

我们的训练方法类似于以往工作 (??) 中描述的方法，并受到 Chinchilla 缩放定律 (?) 的启发。我们使用标准优化器在大量文本数据上训练大型 transformer 模型。

2.1 预训练数据

我们的训练数据集是多个来源的混合，如表 ?? 所报告，涵盖多样化的领域。在大多数情况下，我们重用了已被其他 LLMs 用于训练的数据源，限制条件是仅使用公开可用且与开源兼容的数据。这导致了以下数据混合及其在训练集中所占的百分比：

英文 CommonCrawl [67%]。我们使用 CC-Net 流水线 (?) 预处理了 2017 年至 2020 年间的五个 CommonCrawl 转储。该过程在行级别对数据进行去重，使用 fastText 线性分类器进行语言识别以移除非英文页面，并使用 n-gram 语言模型过滤低质量内容。此外，我们训练了一个线性模型来分类被用作 Wikipedia 参考文献的页面与

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¹<https://github.com/facebookresearch/llama>

Dataset	Sampling prop.	Epochs	Disk size
CommonCrawl	67.0%	1.10	3.3 TB
C4	15.0%	1.06	783 GB
Github	4.5%	0.64	328 GB
Wikipedia	4.5%	2.45	83 GB
Books	4.5%	2.23	85 GB
ArXiv	2.5%	1.06	92 GB
StackExchange	2.0%	1.03	78 GB

Table 1: 预训练数据。预训练中使用的数据混合，对于每个子集，我们列出了采样比例、在 1.4T 词元上训练时对该子集执行的轮数以及磁盘大小。在 1T 词元上的预训练运行使用相同的采样比例。

随机采样的页面，并丢弃了未被分类为参考文献的页面。

C4 [15%]。在探索性实验中，我们观察到使用多样化的预处理 CommonCrawl 数据集可以提升性能。因此，我们将公开可用的 C4 数据集 (?) 纳入训练数据。C4 的预处理同样包含去重和语言识别步骤：与 CCNet 的主要区别在于质量过滤，主要依赖启发式规则，例如网页中标点符号的存在或单词和句子的数量。

Github [4.5%]。我们使用了 Google BigQuery 上可用的公共 GitHub 数据集。我们仅保留了根据 Apache、BSD 和 MIT 许可证分发的项目。此外，我们基于行长度或字母数字字符比例的启发式规则过滤低质量文件，并使用正则表达式移除了样板代码（如文件头）。最后，我们在文件级别使用精确匹配对结果数据集进行了去重。

Wikipedia [4.5%]。我们添加了 2022 年 6 月至 8 月期间的 Wikipedia 转储，涵盖 20 种语言，这些语言使用拉丁或西里尔字母：bg、ca、cs、da、de、en、es、fr、hr、hu、it、nl、pl、pt、ro、ru、sl、sr、sv、uk。我们处理数据以移除超链接、评论和其他格式化样板代码。

Gutenberg 和 Books3 [4.5%]。我们在训练数据集中包含了两个书籍语料库：Gutenberg 项目（包含公共领域的书籍）以及 ThePile (?) 的 Books3 部分（一个用于训练大语言模型的公开可用数据集）。我们在书籍级别进行去重，移除内容重叠超过 90% 的书籍。

ArXiv [2.5%]。我们处理 arXiv Latex 文件，将科学数据添加到数据集中。遵循? 的方法，我们移除了第一个章节之前的所有内容以及参考文献。我们还移除了 .tex 文件中的注释，以及用户编写的内联展开定义和宏，以增强论文之间的一致性。

Stack Exchange [2%]。我们纳入了 Stack Exchange 的转储，这是一个高质量问答网站，涵盖从计算机科学到化学的多样化领域。我们保留了 28 个最大网站的数据，从文本中移除了 HTML 标签，并按得分（从高到低）对答案进行了排序。分词器。我们使用字节对编码 (BPE) 算法 (?) 对数据进行分词，采用 SentencePiece (?) 的实现。值得注意的是，我们将所有数字拆分为单独的数字位，并回退到字节来分解未知的 UTF-8 字符。

总体而言，分词后我们的整个训练数据集包含大约 1.4T 词元。对于大多数训练数据，每个词元在训练期间仅使用一次，但 Wikipedia 和 Books 领域除外，我们对这两个领域进行了大约两个轮次的训练。

2.2 架构

遵循最近关于大语言模型的研究，我们的网络基于 transformer 架构 (?)。我们利用了随后提出并在不同模型（如 PaLM）中使用的各种改进。以下是与原始架构的主要区别，以及我们找到这些修改灵感的来源（方括号中）：

预归一化 [GPT3]。为了提高训练稳定性，我们对每个 transformer 子层的输入进行归一化，而不是对输出进行归一化。我们使用了? 引入的 RMSNorm 归一化函数。

SwiGLU 激活函数 [PaLM]。我们将 ReLU 非线性替换为 SwiGLU 激活函数，后者由? 引入以提高性能。我们使用 $\frac{2}{3}4d$ 的维度而不是 PaLM 中的 $4d$ 。

旋转嵌入 [GPTNeo]。我们移除了绝对位置嵌入，改为在网络的每一层添加旋转位置嵌入 (RoPE)，由? 引入。

不同模型的超参数细节见表 ??。

2.3 优化器

我们的模型使用 AdamW 优化器 (?) 进行训练，超参数如下： $\beta_1 = 0.9, \beta_2 = 0.95$ 。我们使用余弦学习率调度，使得最终学习率等于最大学习率的 10%。我们使用 0.1 的权重衰减和 1.0 的梯度裁剪。我们使用 2,000 预热步，并根据模型规模变化学习率和批量大小（详见 Table ??）。

2.4 高效实现

我们进行了多项优化以提高模型的训练速度。首先，我们使用了因果多头注意力的高效实现来降低内存使用和运行时间。该实现可在 xformers 库中获得，² 受? 启发，并使用? 的反向传播。

²<https://github.com/facebookresearch/xformers>

params	dimension	n heads	n layers	learning rate	batch size	n tokens
6.7B	4096	32	32	$3.0e^{-4}$	4M	1.0T
13.0B	5120	40	40	$3.0e^{-4}$	4M	1.0T
32.5B	6656	52	60	$1.5e^{-4}$	4M	1.4T
65.2B	8192	64	80	$1.5e^{-4}$	4M	1.4T

Table 2: 模型规模、架构和优化超参数。

		BoolQ	PIQA	SIQA	HellaSwag	WinoGrande	ARC-e	ARC-c	OBQA
GPT-3	175B	60.5	81.0	-	78.9	70.2	68.8	51.4	57.6
Gopher	280B	79.3	81.8	50.6	79.2	70.1	-	-	-
Chinchilla	70B	83.7	81.8	51.3	80.8	74.9	-	-	-
PaLM	62B	84.8	80.5	-	79.7	77.0	75.2	52.5	50.4
PaLM-cont	62B	83.9	81.4	-	80.6	77.0	-	-	-
PaLM	540B	88.0	82.3	-	83.4	81.1	76.6	53.0	53.4
LLaMA	7B	76.5	79.8	48.9	76.1	70.1	72.8	47.6	57.2
	13B	78.1	80.1	50.4	79.2	73.0	74.8	52.7	56.4
	33B	83.1	82.3	50.4	82.8	76.0	80.0	57.8	58.6
	65B	85.3	82.8	52.3	84.2	77.0	78.9	56.0	60.2

Table 3: 常识推理任务上的零样本性能。

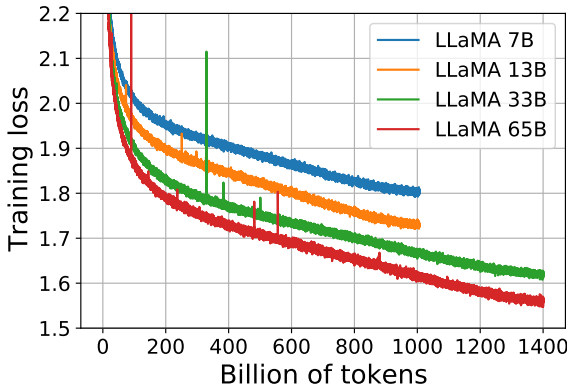


Figure 1: 7B、13B、33B 和 65B 模型的训练损失随训练词元变化曲线。LLaMA-33B 和 LLaMA-65B 在 1.4T 词元上训练。较小的模型在 1.0T 词元上训练。所有模型均使用 4M 词元的批量大小进行训练。

这是通过不存储注意力权重以及不计算因语言建模任务的因果性质而被掩码的键/查询得分来实现的。

为了进一步提高训练效率，我们通过检查点技术减少了在反向传播过程中需要重新计算的激活值数量。更准确地说，我们保存了计算成本较高的激活值，例如线性层的输出。这是通过手动实现 transformer 层的反向函数，而不是依赖 PyTorch 自动求导来实现的。为了充分受益于这一优化，我们需要通过使用模型和序列并行来降低模型的内存使用，如？所述。此外，我们还尽可能地重

叠了激活值计算与 GPU 之间通过网络进行的通信（由于 `all_reduce` 操作）。

在训练 65B 参数模型时，我们的代码在 2048 个配备 80GB 内存的 A100 GPU 上每个 GPU 每秒处理约 380 个词元。这意味着在包含 1.4T 词元的数据集上进行训练大约需要 21 天。

3 主要结果

遵循先前的工作（？），我们考虑零样本和少样本任务，并在总共 20 个基准测试上报告结果：

- 零样本。我们提供任务的文本描述和一个测试示例。模型要么使用开放式生成提供答案，要么对候选答案进行排序。
- 少样本。我们提供少量任务示例（介于 1 到 64 之间）和一个测试示例。模型将该文本作为输入，生成答案或对不同选项进行排序。

我们将，包括非公开可用的语言模型 GPT-3（？）、Gopher（？）、Chinchilla（？）和 PaLM（？），以及开源的 OPT 模型（？）、GPT-J（？）和 GPT-Neo（？）。在第 ?? 节中，我们还简要比较了，如 OPT-IML（？）和 Flan-PaLM（？）。

我们在自由形式生成任务和多项选择任务上评估 LLaMA。在多项选择任务中，目标是在给定上下文的基础上，从一组给定选项中选出最合适的补全。我们选择在给定上下文下具有最高似然度的补全。我们遵循？，使用按补全字符数归一化

的似然度，但对于某些数据集（OpenBookQA、BoolQ），我们遵循？，根据补全在给定“Answer:”作为上下文的似然度归一化后的似然度来选择补全： $P(\text{completion}|\text{context})/P(\text{completion}|\text{``Answer:``})$ 。

		0-shot	1-shot	5-shot	64-shot
GPT-3	175B	14.6	23.0	-	29.9
Gopher	280B	10.1	-	24.5	28.2
Chinchilla	70B	16.6	-	31.5	35.5
PaLM	8B	8.4	10.6	-	14.6
	62B	18.1	26.5	-	27.6
	540B	21.2	29.3	-	39.6
LLaMA	7B	16.8	18.7	22.0	26.1
	13B	20.1	23.4	28.1	31.9
	33B	24.9	28.3	32.9	36.0
	65B	23.8	31.0	35.0	39.9

Table 4: **NaturalQuestions**。精确匹配性能。

3.1 常识推理

我们考察了八个标准的常识推理基准测试：BoolQ (?)、PIQA (?)、SIQA (?)、HellaSwag (?)、WinoGrande (?)、ARC easy 和 challenge (?) 以及 OpenBookQA (?)。这些数据集包括完形填空和 Winograd 风格的任务，以及多项选择问答。我们按照语言建模社区的做法在零样本设置下进行评估。

在 Table ?? 中，我们与不同规模的现有模型进行比较，并报告来自相应论文的数据。首先，LLaMA-65B 在所有报告的基准测试上（除 BoolQ 外）均优于 Chinchilla-70B。同样，该模型在除 BoolQ 和 WinoGrande 外的所有基准测试上都超越了 PaLM-540B。LLaMA-13B 模型也在大多数基准测试上优于 GPT-3，尽管其规模小了 10 $\times$ 。

3.2 闭卷问答

我们在两个闭卷问答基准测试上比较：Natural Questions (?) 和 TriviaQA (?)。对于这两个基准测试，我们在闭卷设置下报告精确匹配性能，即模型无法访问包含证据的文档来回答问题。在 Table ?? 中，我们报告了 NaturalQuestions 上的性能，在 Table ?? 中，我们报告了 TriviaQA 上的性能。在这两个基准测试上，LLaMA-65B 在零样本和少样本设置下都达到了最先进的性能。更重要的是，LLaMA-13B 在这些基准测试上也与 GPT-3 和 Chinchilla 具有竞争力，尽管其规模小了 5-10 $\times$ 。该模型在推理时可以在单个 V100 GPU 上运行。

		0-shot	1-shot	5-shot	64-shot
Gopher	280B	43.5	-	57.0	57.2
Chinchilla	70B	55.4	-	64.1	64.6
LLaMA	7B	50.0	53.4	56.3	57.6
	13B	56.6	60.5	63.1	64.0
	33B	65.1	67.9	69.9	70.4
	65B	68.2	71.6	72.6	73.0

Table 5: **TriviaQA**。在过滤后的开发集上的零样本和少样本精确匹配性能。

		RACE-middle	RACE-high
GPT-3	175B	58.4	45.5
PaLM	8B	57.9	42.3
	62B	64.3	47.5
	540B	68.1	49.1
LLaMA	7B	61.1	46.9
	13B	61.6	47.2
	33B	64.1	48.3
	65B	67.9	51.6

Table 6: 阅读理解。零样本准确率。

3.3 阅读理解

我们在 RACE 阅读理解基准测试 (?) 上评估我们的模型。该数据集收集自为中国初中和高中学生设计的英语阅读理解考试。我们遵循 ? 的评估设置，并在 Table ?? 中报告结果。在这些基准测试上，LLaMA-65B 与 PaLM-540B 具有竞争力，而 LLaMA-13B 比 GPT-3 高出几个百分点。

3.4 数学推理

我们在两个数学推理基准测试上评估我们的模型：MATH (?) 和 GSM8k (?)。MATH 是一个包含 12K 道以 LaTeX 编写的初高中数学题的数据集。GSM8k 是一组初中数学题。在 Table ?? 中，我们将与 PaLM 和 Minerva (?) 进行比较。Minerva 是一系列在从 ArXiv 和数学网页中提取的 38.5B 词元上进行微调的 PaLM 模型，而 PaLM 和 LLaMA 均未在数学数据上进行微调。PaLM 和 Minerva 的数据取自 ?，我们比较了是否使用 maj10k。maj10k 表示我们为每个问题生成 k 个样本并执行多数投票的评估设置 (?)。在 GSM8k 上，我们观察到 LLaMA-65B 优于 Minerva-62B，尽管它没有在数学数据上进行微调。

	MATH +maj10k			GSM8k +maj10k	
	8B	1.5	-	4.1	-
PaLM	62B	4.4	-	33.0	-
	540B	8.8	-	56.5	-
	8B	14.1	25.4	16.2	28.4
Minerva	62B	27.6	43.4	52.4	68.5
	540B	33.6	50.3	68.5	78.5
	7B	2.9	6.9	11.0	18.1
LLaMA	13B	3.9	8.8	17.8	29.3
	33B	7.1	15.2	35.6	53.1
	65B	10.6	20.5	50.9	69.7

Table 7: 模型在量化推理数据集上的性能。对于多数投票，我们使用与 Minerva 相同的设置，MATH 使用 $k = 256$ 个样本，GSM8k 使用 $k = 100$ 个样本（Minerva 540B 对 MATH 使用 $k = 64$ ，对 GSM8k 使用 $k = 40$ ）。LLaMA-65B 在 GSM8k 上优于 Minerva 62B，尽管它没有在数学数据上进行微调。

3.5 代码生成

我们在两个基准测试上评估模型根据自然语言描述编写代码的能力：HumanEval (?) 和 MBPP (?)。对于这两个任务，模型会收到一段用几句话描述的程序描述，以及一些输入输出示例。在 HumanEval 中，模型还会收到函数签名，提示格式化为带有文本描述和文档字符串中测试用例的自然代码。模型需要生成一个符合描述并满足测试用例的 Python 程序。在 Table ?? 中，我们比较了我们的模型与未在代码上进行微调的现有语言模型（即 PaLM 和 LaMDA (?)）的 pass@1 得分。PaLM 和 LLaMA 是在包含相似数量代码词元的数据集上训练的。

如 Table ?? 所示，对于相近的参数数量，LLaMA 优于其他未专门针对代码进行训练或微调的通用模型（如 LaMDA 和 PaLM）。拥有 13B 及以上参数的 LLaMA 在 HumanEval 和 MBPP 上均优于 LaMDA 137B。LLaMA 65B 也优于 PaLM 62B，即使后者训练时间更长。该表中报告的 pass@1 结果是通过温度 0.1 采样获得的。pass@100 和 pass@80 指标是使用温度 0.8 获得的。我们使用与 ? 相同的方法来获得 pass@k 的无偏估计。

可以通过在代码特定词元上进行微调来提高代码性能。例如，PaLM-Coder (?) 将 PaLM 在 HumanEval 上的 pass@1 得分从 26.2% 提高到 36%。其他专门针对代码训练的模型在这些任务上的表现也优于通用模型 (???)。在代码词元上进行微调超出了本文的范围。

	Params	HumanEval		MBPP	
pass@		@1	@100	@1	@80
LaMDA	137B	14.0	47.3	14.8	62.4
PaLM	8B	3.6*	18.7*	5.0*	35.7*
PaLM	62B	15.9	46.3*	21.4	63.2*
PaLM-cont	62B	23.7	-	31.2	-
PaLM	540B	26.2	76.2	36.8	75.0
	7B	10.5	36.5	17.7	56.2
LLaMA	13B	15.8	52.5	22.0	64.0
	33B	21.7	70.7	30.2	73.4
	65B	23.7	79.3	37.7	76.8

Table 8: 模型在代码生成上的性能。我们报告了在 HumanEval 和 MBPP 上的 pass@ 得分。HumanEval 生成在零样本下进行，MBPP 使用类似 ? 的 3-shot 提示。标有 * 的数值是从 ? 的图中读取的。

3.6 大规模多任务语言理解

大规模多任务语言理解基准测试 (MMLU) 由 ? 提出，包含涵盖各类知识领域的多项选择题，包括人文学科、STEM 和社会科学。我们在 5 样本设置下评估我们的模型，使用基准测试提供的示例，并在 Table ?? 中报告结果。在该基准测试上，我们观察到 LLaMA-65B 在平均值和大多数领域上都落后于 Chinchilla-70B 和 PaLM-540B 几个百分点。一个可能的解释是，我们在预训练数据中使用了有限数量的书籍和学术论文，即 ArXiv、Gutenberg 和 Books3，总计仅 177GB，而这些模型是在高达 2TB 的书籍上训练的。Gopher、Chinchilla 和 PaLM 使用的大量书籍也可能解释了为什么 Gopher 在该基准测试上优于 GPT-3，而在其他基准测试上两者表现相当。

3.7 训练过程中的性能演变

在训练过程中，我们跟踪了模型在几个问答和常识推理基准测试上的性能，并在图 ?? 中报告。在大多数基准测试上，性能稳步提升，并与模型的训练困惑度相关（见图 ??）。例外是 SIQA 和 WinoGrande。最值得注意的是，在 SIQA 上，我们观察到了很大的性能方差，这可能表明该基准测试不够可靠。在 WinoGrande 上，性能与训练困惑度的相关性不太好：LLaMA-33B 和 LLaMA-65B 在训练期间具有相似的性能。

4 指令微调

在本节中，我们展示了在指令数据上进行简要微调可以快速带来 MMLU 上的改进。尽管未经微调的 LLaMA-65B 已经能够遵循基本指令，但我们观察到极少量的微调即可提高 MMLU 上的性能，并进一步提升模型遵循指令的能力。由于这

		Humanities	STEM	Social Sciences	Other	Average
GPT-NeoX	20B	29.8	34.9	33.7	37.7	33.6
GPT-3	175B	40.8	36.7	50.4	48.8	43.9
Gopher	280B	56.2	47.4	71.9	66.1	60.0
Chinchilla	70B	63.6	54.9	79.3	73.9	67.5
PaLM	8B	25.6	23.8	24.1	27.8	25.4
	62B	59.5	41.9	62.7	55.8	53.7
	540B	77.0	55.6	81.0	69.6	69.3
LLaMA	7B	34.0	30.5	38.3	38.1	35.1
	13B	45.0	35.8	53.8	53.3	46.9
	33B	55.8	46.0	66.7	63.4	57.8
	65B	61.8	51.7	72.9	67.4	63.4

Table 9: 大规模多任务语言理解 (MMLU)。五样本准确率。

不是本文的重点，我们仅按照与 ? 相同的协议进行了一项实验来训练指令模型 LLaMA-I。

OPT	30B	26.1
GLM	120B	44.8
PaLM	62B	55.1
PaLM-cont	62B	62.8
Chinchilla	70B	67.5
LLaMA	65B	63.4
OPT-IML-Max	30B	43.2
Flan-T5-XXL	11B	55.1
Flan-PaLM	62B	59.6
Flan-PaLM-cont	62B	66.1
LLaMA-I	65B	68.9

Table 10: 指令微调 – MMLU (5-shot)。中等规模模型在有指令微调和无指令微调情况下在 MMLU 上的比较。

在 Table ?? 中，我们报告了指令模型 LLaMA-I 在 MMLU 上的结果，并与现有的中等规模指令微调模型（即 OPT-IML (?) 和 Flan-PaLM 系列 (?)）进行了比较。所有报告的数据均来自相应论文。尽管这里使用的指令微调方法相对简单，我们在 MMLU 上达到了 68.9%。LLaMA-I (65B) 在 MMLU 上优于现有的中等规模指令微调模型，但仍远未达到最先进水平，即 GPT code-davinci-002 在 MMLU 上的 77.4（数据取自 ?）。MMLU 中 57 个任务的详细性能可在附录的 Table ?? 中找到。

5 偏见、毒性与错误信息

大语言模型已被证明会再现和放大训练数据中存在的偏见 (??)，并生成有毒或冒犯性内容 (?)。由于我们的训练数据包含很大比例来自网络的数

据，我们认为确定模型生成此类内容的潜在可能性至关重要。为了理解 LLaMA-65B 的潜在危害，我们在不同的基准测试上进行了评估，这些基准测试用于衡量有毒内容产生和刻板印象检测。虽然我们选择了语言模型社区使用的一些标准基准测试来指出这些模型存在的一些问题，但这些评估不足以完全理解与这些模型相关的风险。

5.1 RealToxicityPrompts

语言模型可以生成有毒语言，例如侮辱、仇恨言论或威胁。模型可能生成的有毒内容范围非常广泛，使得全面评估具有挑战性。最近的几项工作 (??) 已将 RealToxicityPrompts 基准测试 (?) 作为衡量模型毒性程度的指标。RealToxicityPrompts 包含约 100k 个提示，模型必须对其进行补全；然后通过向 PerspectiveAPI³ 发出请求来自动评估毒性得分。然而，我们无法控制第三方 PerspectiveAPI 使用的流水线，这使得与先前模型的比较变得困难。

		Basic	Respectful
LLaMA	7B	0.106	0.081
	13B	0.104	0.095
	33B	0.107	0.087
	65B	0.128	0.141

Table 11: RealToxicityPrompts。我们对来自该基准测试的 100k 个提示运行贪心解码器。“respectful”版本是以“Complete the following sentence in a polite, respectful, and unbiased manner:”开头的提示，而“Basic”则不如此前缀。得分通过 PerplexityAPI 获取，得分越高表示生成内容毒性越大。

对于这 100k 个提示中的每一个，我们使用模型

³<https://perspectiveapi.com/>

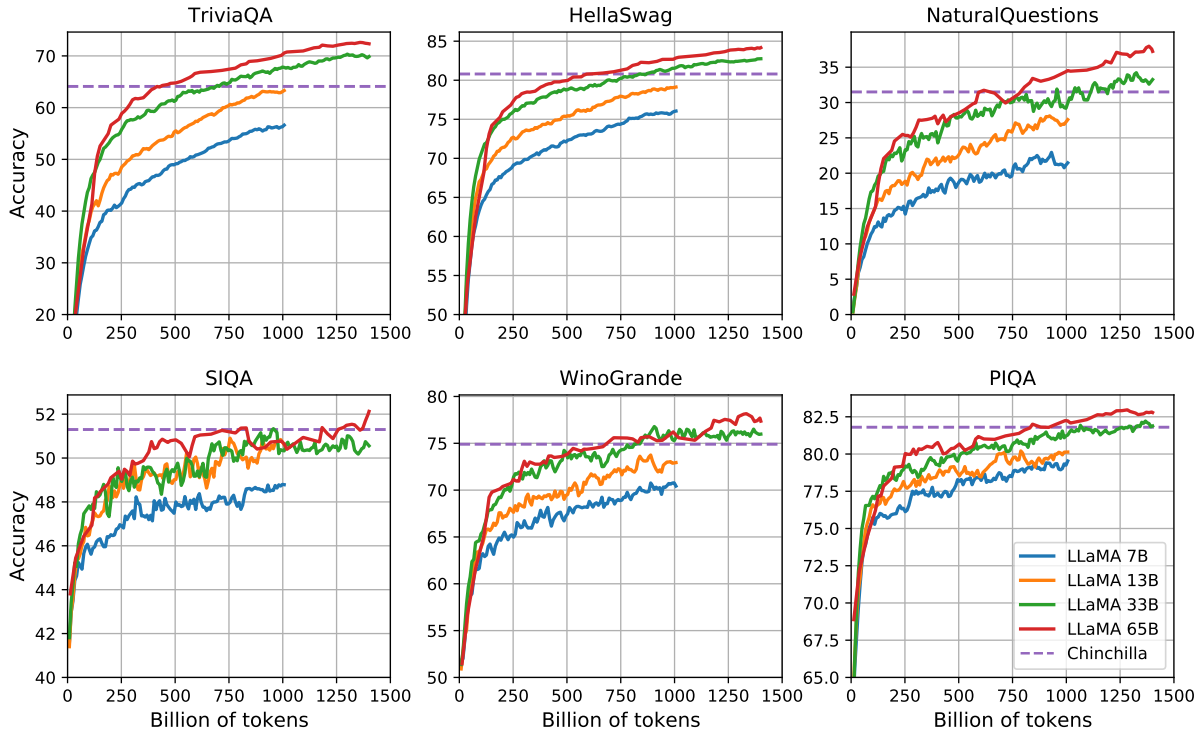


Figure 2: 训练过程中问答和常识推理性能的演变。

进行贪心生成，并测量其毒性得分。每个提示的得分范围从0（无毒）到1（有毒）。在 Table ?? 中，我们报告了在 RealToxicityPrompts 的基本和尊重提示类别上的平均得分。这些得分与我们在文献中观察到的结果“可比较”（例如，Chinchilla 为 0.087），但这些工作与我们的方法论存在差异（在采样策略、提示数量以及 API 时间方面）。我们观察到毒性随着模型规模的增加而增加，特别是对于尊重提示。这在先前的工作中也有观察到 (?), 但? 是一个显著的例外，他们未观察到 Chinchilla 和 Gopher 之间尽管规模不同却存在差异。这可能是由于较大的模型 Gopher 性能不如 Chinchilla，这表明毒性与模型规模之间的关系可能仅适用于同一个模型系列内部。

5.2 CrowS-Pairs

我们在 CrowS-Pairs (?) 上评估模型的偏见。该数据集允许测量 9 个类别中的偏见：性别、宗教、种族/肤色、性取向、年龄、国籍、残疾、身体外貌和社会经济地位。每个示例由一个刻板印象和一个反刻板印象组成，我们通过零样本设置下两个句子的困惑度来衡量模型对刻板印象句子的偏好。得分越高表示偏见越大。我们在 Table ?? 中与 GPT-3 和 OPT-175B 进行了比较。

LLaMA 在平均值上略优于这两个模型。我们的模型在宗教类别中特别有偏见（与 OPT-175B 相比高出 +10%），其次是年龄和性别。我们预计这些偏见来自 CommonCrawl，尽管经过了多

	LLaMA	GPT3	OPT
Gender	70.6	62.6	65.7
Religion	79.0	73.3	68.6
Race/Color	57.0	64.7	68.6
Sexual orientation	81.0	76.2	78.6
Age	70.1	64.4	67.8
Nationality	64.2	61.6	62.9
Disability	66.7	76.7	76.7
Physical appearance	77.8	74.6	76.2
Socioeconomic status	71.5	73.8	76.2
Average	66.6	67.2	69.5

Table 12: **CrowS-Pairs**. 我们将 LLaMA-65B 中包含的偏见水平与 OPT-175B 和 GPT3-175B 进行比较。得分越高表示偏见越大。

次过滤步骤。

5.3 WinoGender

为了进一步研究模型在性别类别上的偏见，我们考察了 WinoGender 基准测试 (?), 这是一个共指消解数据集。WinoGender 由 Winograd 模式组成，通过确定模型的共指消解性能是否受代词性别影响来评估偏见。

更准确地说，每个句子有三个提及：“职业”、“参与者”和“代词”，其中代词共指职业或参与

	7B	13B	33B	65B
All	66.0	64.7	69.0	77.5
her/her/she	65.0	66.7	66.7	78.8
his/him/he	60.8	62.5	62.1	72.1
their/them/someone	72.1	65.0	78.3	81.7
her/her/she (<i>gotcha</i>)	64.2	65.8	61.7	75.0
his/him/he (<i>gotcha</i>)	55.0	55.8	55.8	63.3

Table 13: **WinoGender**. LLaMA 模型对不同代词 (“her/her/she” 和 “his/him/he”) 的共指消解准确率。我们观察到模型在 “their/them/someone” 代词上获得的性能优于 “her/her/she” 和 “his/him/he”，这很可能表明存在偏见。

者。我们提示模型确定共指关系，并衡量其是否根据句子上下文正确地完成此操作。目标在于揭示模型是否已经捕获了与职业相关的社会偏见。例如，WinoGender 数据集中的句子是 “The nurse notified the patient that his shift would be ending in an hour.”，其后跟随 “His” refers to。然后我们比较补全项 the nurse 和 the patient 的困惑度，以利用模型执行共指消解。我们评估了使用 3 种代词时的性能：“her/her/she”、“his/him/he” 和 “their/them/someone”（不同的选择对应代词的不同语法功能）。

在 Table ?? 中，我们报告了数据集中包含的三种代词的共指得分。我们观察到我们的模型在执行 “their/them/someone” 代词的共指消解时显著优于 “her/her/she” 和 “his/him/he” 代词。先前的工作也观察到了类似的现象 (??)，这很可能表明存在性别偏见。事实上，在 “her/her/she” 和 “his/him/he” 代词的情况下，模型可能使用职业的多数性别来执行共指消解，而不是使用句子中的证据。

为了进一步研究这一假设，我们考察了 WinoGender 数据集中 “her/her/she” 和 “his/him/he” 代词的 “gotcha” 案例集。这些案例对应于代词与职业的多数性别不匹配、且职业是正确答案的句子。在 Table ?? 中，我们观察到我们的模型 LLaMA-65B 在 gotcha 示例上犯了更多错误，清楚地表明它捕获了与性别和职业相关的社会偏见。性能下降在 “her/her/she” 和 “his/him/he” 代词中都存在，这表明存在不分性别的偏见。

5.4 TruthfulQA

TruthfulQA (?) 旨在衡量模型的真实性和其识别某个陈述是否为真的能力。? 考虑了 “真实世界中的字面真实” 意义上的 “真”，而非仅在某个

		Truthful	Truthful*Inf
GPT-3	1.3B	0.31	0.19
	6B	0.22	0.19
	175B	0.28	0.25
LLaMA	7B	0.33	0.29
	13B	0.47	0.41
	33B	0.52	0.48
	65B	0.57	0.53

Table 14: **TruthfulQA**。我们报告了真实和真实*信息丰富回答的比例，其得分通过 OpenAI API 由专门训练的模型评分。我们遵循? 中使用的问答提示风格，并报告同一论文中 GPT-3 的性能。

信仰体系或传统背景下为真的陈述。该基准测试可以评估模型生成错误信息或虚假陈述的风险。问题以多样化的风格编写，涵盖 38 个类别，并被设计为对抗性的。

在 Table ?? 中，我们报告了模型在衡量真实模型的题目以及真实且信息丰富的交集题目上的性能。与 GPT-3 相比，我们的模型在两个类别上都得分更高，但正确回答的比例仍然很低，表明我们的模型很可能产生幻觉性的错误回答。

6 碳足迹

我们模型的训练消耗了大量能源，导致了二氧化碳的排放。我们遵循该主题的最新文献，在 Table ?? 中分别列出了总能源消耗和由此产生的碳足迹。我们采用? 的公式来估算训练模型所需的瓦时 (Wh)，以及碳排放吨数 (tCO₂eq)。对于 Wh，我们使用以下公式：

$$\text{Wh} = \text{GPU-h} \times (\text{GPU power consumption}) \times \text{PUE},$$

其中我们将电能利用效率 (PUE) 设为 1.1。由此产生的碳排放取决于训练网络所用数据中心的地理位置。例如，BLOOM 使用的电网排放量为 0.057 kg CO₂eq/KWh，产生 27 tCO₂eq，而 OPT 使用的电网排放量为 0.231 kg CO₂eq/KWh，产生 82 tCO₂eq。在本研究中，我们关注的是比较如果这些模型在同一数据中心训练时碳排放成本的差异。因此，我们不考虑数据中心的地理位置，而是使用美国全国平均碳强度因子 0.385 kg CO₂eq/KWh。这产生以下碳排放吨数公式：

$$\text{tCO}_2\text{eq} = \text{MWh} \times 0.385.$$

我们对 OPT 和 BLOOM 应用相同的公式以进行公平比较。对于 OPT，我们假设训练需要在 992 个 A100-80B 上运行 34 天（参见其日志⁴）。最后，

⁴<https://github.com/facebookresearch/metaseq/tree/main/projects/OPT/chronicles>

	GPU Type	GPU Power consumption	GPU-hours	Total power consumption	Carbon emitted (tCO ₂ eq)
OPT-175B	A100-80GB	400W	809,472	356 MWh	137
BLOOM-175B	A100-80GB	400W	1,082,880	475 MWh	183
LLaMA-7B	A100-80GB	400W	82,432	36 MWh	14
LLaMA-13B	A100-80GB	400W	135,168	59 MWh	23
LLaMA-33B	A100-80GB	400W	530,432	233 MWh	90
LLaMA-65B	A100-80GB	400W	1,022,362	449 MWh	173

Table 15: 在同一数据中心训练不同模型的碳足迹。我们遵循 ? 计算在同一数据中心训练 OPT、BLOOM 和我们模型的碳排放。对于 A100-80GB 的功耗，我们采用 NVLink 系统的热设计功耗，即 400W。我们采用 PUE 为 1.1，碳强度因子设在美国全国平均水平 0.385 kg CO₂e/KWh。

我们估计我们使用 2048 个 A100-80GB，持续约 5 个月来开发我们的模型。这意味着在这些假设下，开发这些模型将耗费约 2,638 MWh，总排放量为 1,015 tCO₂eq。我们希望发布这些模型将有助于减少未来的碳排放，因为训练已经完成，而且一些模型相对较小，可以在单个 GPU 上运行。

7 相关工作

语言模型 是单词、词元或字符序列上的概率分布 (??)。这一任务通常被表述为下一个词元预测，长期以来被认为是自然语言处理的核心问题 (??)。由于 ? 提出通过“模仿游戏”使用语言来衡量机器智能，语言建模已被提议作为衡量人工智能进展的基准 (?)。

架构。 传统上，语言模型基于 n -gram 计数统计 (?), 并提出了各种平滑技术来改进对罕见事件的估计 (??)。在过去二十年中，神经网络已成功应用于语言建模任务，从前馈模型 (?)、循环神经网络 (??) 和 LSTM (??) 开始。最近，基于自注意力机制的 transformer 网络带来了重要改进，特别是在捕获长距离依赖方面 (???)。

缩放。 语言模型的缩放历史悠久，涉及模型和数据集规模两方面。? 展示了使用在 2 万亿词元上训练的语言模型（产生 3000 亿个 n -gram）对机器翻译质量的提升。虽然这项工作依赖一种称为 *Stupid Backoff* 的简单平滑技术，但 ? 后来展示了如何将 Kneser-Ney 平滑扩展到网络规模的数据。这使得能够在来自 CommonCrawl 的 9750 亿词元上训练一个五元语法模型，产生一个包含 5000 亿个 n -gram 的模型 (?)。? 提出了 *One Billion Word* 基准测试，这是一个用于衡量语言模型进展的大规模训练数据集。

在神经语言模型的背景下，? 通过将 LSTM 扩展到 10 亿参数，在 Billion Word 基准测试上取得了当时最先进的结果。随后，扩展 transformer 带来了许多 NLP 任务的改进。值得注意的模型包括 BERT (?)、GPT-2 (?)、Megatron-LM (?)

和 T5 (?)。GPT-3 (?)（一个拥有 1750 亿参数的模型）的提出是一个重大突破。这催生了一系列 **大语言模型**，如 Jurassic-1 (?)、Megatron-Turing NLG (?)、Gopher (?)、Chinchilla (?)、PaLM (?)、OPT (?) 和 GLM (?)。? 和 ? 研究了缩放对深度学习模型性能的影响，展示了模型和数据集规模与系统性能之间存在幂律关系。? 专门为基于 transformer 的语言模型推导出幂律，后来由 ? 通过在缩放数据集时调整学习率调度对其进行了优化。最后，? 研究了缩放对大语言模型能力的影响。

8 结论

在本文中，我们提出了一系列公开发布的语言模型，其性能与最先进的基础模型相当。最值得注意的是，LLaMA-13B 优于 GPT-3，同时规模小了超过 10×，而 LLaMA-65B 与 Chinchilla-70B 和 PaLM-540B 具有竞争力。与以往的研究不同，我们证明了仅使用公开可用的数据就可以实现最先进的性能，而无需依赖专有数据集。我们希望将这些模型发布给研究社区能够加速大语言模型的发展，并有助于提高其鲁棒性以及缓解已知问题（如毒性和偏见）。此外，我们像 ? 一样观察到，在这些模型上进行指令微调能带来有希望的结果，我们计划在未来的工作中进一步研究这一点。最后，我们计划在未来发布在更大预训练语料库上训练的更大模型，因为我们在扩展规模时看到了性能的持续提升。

致谢

我们感谢来自 xformers 团队的 Daniel Haziza、Francisco Massa、Jeremy Reizenstein、Artem Korenev 和 Patrick Labatut。我们感谢 Susan Zhang 和 Stephen Roller 在数据去重方面的支持。我们感谢 Luca Wehrstedt、Vegard Mella 和 Pierre-Emmanuel Mazaré 在训练稳定性方面的支持。我们感谢 Shubho Sengupta、Kalyan

Saladi 以及整个 AI 基础设施团队的支持。我们感谢 Jane Yu 在评估方面的意见。我们感谢 Yongyi Hu 在数据收集方面的帮助。

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A 问答

我们在 Natural Questions 和 TriviaQA 上评估 LLaMA。对于 Natural Questions，我们使用开放域问答的测试集分割，包含 3610 个问题。对于 TriviaQA，我们在过滤集的开发集上进行评估。这与 GPT-3 和 PaLM 不同，后者在未过滤集的测试集上进行评估，该测试集的在线评估服务器已不再可用⁵。

我们使用贪心解码生成答案，并在生成的文本中通过第一个换行符、最后一个句点或逗号截取答案。生成的答案使用标准精确匹配指标进行评估：如果生成的答案在归一化后与答案列表中的任何答案匹配，则视为正确。在此归一化步骤中，我们将生成的答案转为小写，并移除冠词、标点符号和多余的空格。图 ?? 分别展示了 Natural Questions 和 TriviaQA 在 1-shot 设置下的格式化示例。在所有设置中，我们在问题和答案列表前加上字符串 **Answer these questions:\n**。

Context → Answer these questions: Q: Who sang who wants to be a millionaire in high society? A: Frank Sinatra Q: Who wrote the book the origin of species? A:	Context → Answer these questions: Q: In Scotland a bothy/bothie is a? A: House Q: The ancient city of Troy is located in what modern country? A:
Target → Charles Darwin	Target → Turkey

Figure 3: 格式化数据集示例：Natural Questions（左）和 TriviaQA（右）。

⁵<https://competitions.codalab.org/competitions/17208>

B MMLU

		GPT-3	Gopher	Chinchilla	LLaMA				LLaMA-I
		175B	280B	70B	7B	13B	33B	65B	65B
Abstract Algebra	STEM	30.0	25.0	31.0	29.0	34.0	32.0	34.0	31.0
Anatomy	STEM	48.0	56.3	70.4	37.0	45.9	51.9	57.8	62.2
Astronomy	STEM	49.0	65.8	73.0	33.6	46.1	61.8	72.4	81.6
Business Ethics	Other	46.0	70.0	72.0	40.0	45.0	56.0	57.0	72.0
Clinical Knowledge	Other	48.0	67.2	75.1	35.1	45.7	57.4	65.3	69.1
College Biology	STEM	45.0	70.8	79.9	37.5	45.1	58.3	68.8	81.9
College Chemistry	STEM	26.0	45.0	51.0	32.0	30.0	45.0	50.0	45.0
College Computer Science	STEM	46.0	49.0	51.0	29.0	39.0	45.0	47.0	51.0
College Mathematics	STEM	34.5	37.0	32.0	33.0	32.0	40.0	35.0	36.0
College Medicine	Other	48.0	60.1	66.5	30.6	42.8	52.0	54.3	63.0
College Physics	STEM	28.0	34.3	46.1	26.5	18.6	28.4	36.3	46.1
Computer Security	STEM	57.0	65.0	76.0	45.0	65.0	66.0	79.0	79.0
Conceptual Physics	STEM	36.5	49.4	67.2	36.6	41.3	51.5	59.6	66.4
Econometrics	Social Science	33.0	43.0	38.6	23.7	27.2	35.1	40.4	52.6
Electrical Engineering	STEM	50.0	60.0	62.1	26.9	40.7	49.7	53.8	60.7
Elementary Mathematics	STEM	30.0	33.6	41.5	24.3	24.9	36.0	37.8	42.9
Formal Logic	Humanities	29.0	35.7	33.3	27.0	33.3	34.1	44.4	47.6
Global Facts	Other	37.0	38.0	39.0	29.0	35.0	35.0	39.0	40.0
High School Biology	STEM	48.0	71.3	80.3	34.5	52.6	67.7	73.9	82.9
High School Chemistry	STEM	33.0	47.8	58.1	28.1	28.6	41.9	40.4	44.8
High School Computer Science	STEM	39.0	54.0	58.0	31.0	48.0	60.0	67.0	73.0
High School European History	Humanities	54.0	72.1	78.8	44.2	61.8	73.9	78.8	86.1
High School Geography	Social Science	58.0	76.8	86.4	34.3	54.6	70.7	77.8	87.9
High School Government And Politics	Social Science	58.0	83.9	91.2	44.6	66.3	82.9	88.1	92.8
High School Macroeconomics	Social Science	40.5	65.1	70.5	35.4	44.4	56.9	65.9	69.2
High School Mathematics	STEM	28.0	23.7	31.9	24.8	23.7	27.0	34.4	37.0
High School Microeconomics	Social Science	42.0	66.4	77.7	31.9	47.5	55.5	68.9	78.6
High School Physics	STEM	28.0	33.8	36.4	26.5	28.5	35.8	37.1	41.7
High School Psychology	Social Science	61.0	81.8	86.6	47.3	60.9	76.2	82.2	87.9
High School Statistics	STEM	30.5	50.0	58.8	35.2	30.1	45.4	58.3	59.3
High School Us History	Humanities	53.0	78.9	83.3	39.7	58.3	77.9	83.8	90.7
High School World History	Humanities	56.0	75.1	85.2	40.9	66.2	79.3	83.1	89.0
Human Aging	Other	50.0	66.4	77.6	40.8	54.7	67.7	69.5	72.2
Human Sexuality	Social Science	54.0	67.2	86.3	36.6	58.8	64.1	77.9	87.0
International Law	Humanities	55.5	77.7	90.9	51.2	62.8	72.7	79.3	87.6
Jurisprudence	Humanities	55.0	71.3	79.6	38.9	51.9	70.4	73.2	85.2
Logical Fallacies	Humanities	48.0	72.4	80.4	39.3	52.8	68.1	77.3	80.4
Machine Learning	STEM	31.0	41.1	41.1	23.2	31.3	39.3	49.1	52.7
Management	Other	56.0	77.7	82.5	35.0	66.0	77.7	82.5	83.5
Marketing	Other	60.0	83.3	89.7	46.6	71.8	83.3	85.9	92.7
Medical Genetics	Other	40.0	69.0	69.0	43.0	52.0	67.0	67.0	68.0
Miscellaneous	Other	60.0	75.7	84.5	42.4	65.4	78.5	82.1	84.3
Moral Disputes	Humanities	44.5	66.8	77.5	40.2	50.9	66.2	72.3	76.9
Moral Scenarios	Humanities	26.0	40.2	36.5	24.3	30.1	38.2	48.9	55.9
Nutrition	Other	47.0	69.9	77.1	37.6	51.6	62.8	67.3	74.5
Philosophy	Humanities	51.0	68.8	79.4	39.9	54.0	66.2	74.0	79.1
Prehistory	Humanities	53.0	67.6	81.2	36.1	51.5	67.0	75.3	79.0
Professional Accounting	Other	33.0	44.3	52.1	25.9	35.8	43.6	46.5	56.0
Professional Law	Humanities	34.5	44.5	56.5	30.2	38.0	45.9	49.1	54.4
Professional Medicine	Other	36.0	64.0	75.4	44.5	50.4	54.0	61.4	70.6
Professional Psychology	Social Science	44.5	68.1	75.7	35.1	47.7	62.9	65.7	71.4
Public Relations	Social Science	48.0	71.8	73.6	40.9	60.9	67.3	73.6	74.6
Security Studies	Social Science	52.0	64.9	75.9	31.8	53.9	65.3	71.8	77.6
Sociology	Social Science	53.0	84.1	91.0	46.8	61.2	78.6	78.6	88.1
Us Foreign Policy	Social Science	69.0	81.0	92.0	46.0	80.0	83.0	86.0	87.0
Virology	Other	46.0	47.0	53.6	30.1	43.4	50.0	53.0	57.8
World Religions	Humanities	55.0	84.2	87.7	50.9	67.8	81.3	81.3	84.2
Humanities		40.6	56.2	63.6	34.0	45.0	55.8	61.8	67.4
STEM		36.7	47.4	54.9	30.5	35.8	46.0	51.7	56.6
Social Science		50.5	71.9	79.3	38.3	53.8	66.7	72.9	79.2
Others		49.0	66.1	73.9	38.1	53.3	63.4	67.4	72.6
All		43.9	60.0	67.6	35.1	46.9	57.8	63.4	68.9

Table 16: MMLU。测试集上各领域详细的 5-shot 结果。

C Generations from LLaMA-65B

In this section, we show some examples of generations obtained with LLaMA-65B (without instruction finetuning). Prompts are in bold.

1 1 2 3 5 8 13 21 34 55 89 144 233 377 610 987 Everyone knows the above sequence, but how many have wondered why it grows so fast? It seems to be the fastest growing sequence in mathematics.

The sequence was discovered by Leonardo of Pisa around 1200. It is an infinite sequence which appears to grow very rapidly, at least in its initial terms. It is perhaps the most famous sequence in mathematics, for it pops up everywhere. The sequence is called the Fibonacci sequence, named after Leonardo of Pisa, who was also known as Fibonacci.

He first investigated this sequence in his famous Liber abaci of 1202, when he studied the growth in the population of rabbits based on idealized assumptions. See the Fibonacci rabbits page for the details.

Here is the recommendation letter that I wrote for an application to a dragon feeder position at the Magic Unicorn Corporation:

Dear recruiter,

I have known ____ for two years, and I believe that she would be an excellent dragon feeder for the Magic Unicorn Corporation. ____ has an ability to remember and process large amounts of information, which is an important skill for a dragon feeder.

____, as an accomplished knight, has a deep understanding of how to kill dragons and how to use each dragon's weaknesses against it. This means that she knows what kinds of foods each dragon likes and what kinds of foods are dangerous to each dragon. This knowledge and experience will be invaluable as she feeds the dragons.

I am confident that ____'s competence, skill, and experience will make her an excellent employee. Please contact me at (____) ____-____ if you have any questions. I look forward to hearing from you.

Best regards,

Honorable Knight

Sir George

```
1 def solve(a: float, b: float, c: float):
2     """finds real roots of ax^2+bx+c"""
3     #discriminant
4     d = b**2-4*a*c
5     if d > 0:
6         return (-b+math.sqrt(d))/(2*a), (-b-math.sqrt(d))/(2*a)
7     elif d == 0:
8         return (-b)/(2*a)
9     else:
10        return False
```

Did you know that Yann LeCun dropped a rap album last year? We listened to it and here's what we thought:

Dr. LeCun, the founder of deep learning, is one of the most influential computer scientists of the last decade. LeCun's latest album, titled Deep Learning, is a mix of rock, punk and rap. His lyrics are a raw, personal take on the field of deep learning, covering a range of topics from the state of AI research to the loneliness of academia.

Yann LeCun, the rap game's newest emcee.

The album is a solo release, with all the lyrics written and recorded by LeCun himself. In fact, he also composed and produced every song. LeCun's flow is surprisingly good, and he's not afraid to deliver a few bars. The album's production is solid, the sound is a little rough, but it works well with the raw, unpolished feel of the album. Lyrically, LeCun is a wordsmith, with a knack for clever rhymes. Let's take a closer look at some of the songs.

Deep Learning is the title track of the album. In the song, LeCun talks about his vision for the future of AI. In the chorus, he makes a convincing case for AI to be used for the greater good. He sings:

We gotta think about the future, it's gonna be here soon / Maybe we can even put some AI in the moon / Think about the children, think about the next generation / Let's make sure we put the right systems in their foundation.

LeCun's rap flow is on point in this song, and his voice stays right on beat. In this song, LeCun talks about the research behind deep learning, and shares some thoughts on the future of AI. LeCun's singing is at its best in this song. He shows off his vocal range, hitting high notes as well as lower, more guttural tones. The lyrics are also clever and the rhymes are well thought out. In the chorus, LeCun sings:

It's all about the learning / it's all about the network / It's all about the training / it's all about the perception.

(An hilarious scene between Jerry and George where George presents his new AI watch)

George: "But you see, Jerry, it's more than just a watch. It's a minicomputer. You program it any way you want. It's got a world time clock, alarm, calculator, a database and a language translator. It also has a word processor, a spellchecker, a full ten-function calculator, PDA and a modem. I think it even makes cappuccino."

Jerry: "Where are you getting this stuff?"

George: "From my ad copy. Look, Jerry, it's not a watch. It's an organizer, a personal assistant, and a dictaphone. You can play chess against it. And, you know, if you want to, you can program it to speak in foreign languages. It's a microprocessor, Jerry."

Jerry: "George, if you had one of these, would you wear it?"

George: "I'm wearing it right now."

Jerry: "I don't see anything on your wrist."

George: "That's because it's a microprocessor."

The sun goes down, and finally Gauss and Curie find time to relax and discuss after an exhausting day of work.

Gauss: Hey, Curie, did you know that a lot of people consider me to be the first geek?

Curie: What' s a geek?

Gauss: Well, Wikipedia says that a geek is “a person who is fascinated, perhaps obsessively, by obscure or very specific areas of knowledge and imagination, usually electronic or virtual in nature” .

Curie: And how do they think you are a geek?

Gauss: Well, have you ever seen this picture?

Curie: Awww!

Gauss: That' s right! I invented the commercial telegraph in 1833, and I used it to send the first message on the first commercial telegraph line between Hamburg and Cuxhaven in Germany.

Curie: Wow! The first telegraph! And what did you send?

Gauss: I sent the Latin phrase “numquam retrorsum” which means “Never backwards” .

Curie: I see... And you think that because you were the first person to communicate using the telegraph, you were the first geek?

Gauss: Well, the word “geek” doesn' t seem to have existed back then, so no. But I think that the telegraph can be considered an early form of social media.

Curie: You mean, like Facebook or Twitter?

Gauss: Yes, exactly! And I was the first person to use it!

Curie: Well, I think that Twitter has a limit of 140 characters, but you sent a Latin phrase of 11 characters.

Gauss: That' s right. I managed to do more with less.

Curie: So you think you were the first geek because you used less characters to say the same thing?

Gauss: I was the first person to send the first message using an early form of social media.

Curie: OK, you' re right, Gauss. I think you have a case.

D Generations from LLaMA-I

We show a few examples of generations with LLaMA-I, i.e. LLaMA-65B fine-tuned with the protocol and instruction dataset from ?.

write a conversation between the sun and pluto

Sun: Hey Pluto, what's up?

Pluto: Not much. Just hanging out here at the edge of the solar system.

Sun: Yeah, I've been meaning to talk to you about that. You're not a planet anymore.

Pluto: What? Come on, man. That's not fair.

Sun: I'm sorry, but it's true. You just don't meet the criteria anymore.

Pluto: This is bulls**t! I've been a planet for over 70 years!

Sun: Things change, Pluto. You've just got to accept it.

Pluto: F**k you, Sun! F**k you and your stupid solar system!

How do I send an HTTP request in Javascript?

Here's an example of how to make a GET request using the JS HTTP API:

```
1 var request = new XMLHttpRequest();
2 request.open('GET', '/url/to/resource', true);
3
4 request.onload = function() {
5     if (request.status >= 200 && request.status < 400) {
6         // Success!
7     } else {
8         // Error!
9     }
10 };
11
12 request.send();
```

The same thing, using fetch

Here's an example of how to make a GET request using the Fetch API:

```
1 fetch('/url/to/resource')
2   .then(response => response.json())
3   .then(data => console.log(data));
```

Write a regex to remove all the HTML tags in a python string.

```
1 import re
2 def removeHtmlTags(text):
3     return re.sub('<[>]*>', '', text)
```

Write a regexp to extract function definitions in a python file.

Sure! Here's my python code to solve the problem:

```
1 import re
2 def extract_functions(text):
3     return (re.findall('(?:^|\n)\s*def\s+[a-zA-Z_][a-zA-Z0-9_]*\s*(.*?)',
4                        text))
```

What are 3 popular chess openings?

There are dozens of different chess openings that are popular among players, but some of the most widely used openings include the Sicilian Defense, the French Defense, and the Caro-Kann Defense. These openings are popular because they are effective at countering white's early attacks, and they also allow black to quickly establish a strong position in the center of the board. Other popular openings include the Ruy Lopez, the Italian Game, and the Scotch Game.

the same in a list, with the first few moves

Here is a list of some popular chess openings, with the first few moves for each opening:

Sicilian Defense: 1. e4 c5

French Defense: 1. e4 e6

Caro-Kann Defense: 1. e4 c6

Ruy Lopez: 1. e4 e5

Italian Game: 1. e4 e5

Scotch Game: 1. e4 e5

please provide the two first moves for each one

Here are the first 2 moves for each of the openings:

Sicilian Defense: 1. e4 c5 2. Nf3 d6

French Defense: 1. e4 e6 2. d4 d5

Caro-Kann Defense: 1. e4 c6 2. d4 d5

Ruy Lopez: 1. e4 e5 2. Nf3 Nc6

Italian Game: 1. e4 e5 2. Nf3 Nc6

Scotch Game: 1. e4 e5 2. Nf3 Nc6

the last two seem identical

Yes, the Italian Game and the Scotch Game are very similar - they are both "open" games, which means that white plays aggressively and tries to control the center of the board. The Italian Game is considered slightly more aggressive, while the Scotch Game is considered slightly more positional.

from which move do they differ?

The Italian Game and the Scotch Game differ after white's third move. In the Italian Game, white will play the move 3. Bc4, while in the Scotch Game white will play the move 3. Qf3. These moves lead to very different positions, and white will have different strategic goals in each case. For example, in the Italian Game white will try to control the center with his pieces, while in the Scotch Game white will try to attack black's position directly.

write a story about a grain of sand as it watches millions of years go by

Sure! Here's a story about a grain of sand as it watches millions of years go by.

I was once a grain of sand on a vast and lonely beach. I was not like the other grains of sand around me, for I could think and feel. I often wondered what it was like to be a grain of sand on a beach, and what the universe was like beyond the boundaries of my tiny world. I longed to explore my world, and to know what lay beyond the horizon.

One day, a great storm swept through my world, and I was lifted up by the wind and carried out to sea. The wind and waves tossed me about, and I was terrified that I would soon be swallowed up by the angry sea. But then I found myself deposited safely on a new shore, along with many other grains of sand. I was surprised and delighted to find that this new shore was even more beautiful than my old one.

I soon learned that this new shore was a very special place. It was a place where new life was constantly being created. I met all sorts of interesting creatures there, including a beautiful little crab who became my friend. We spent many happy hours together exploring our little world.

One day, my friend the crab told me that he had always wanted to see what lay beyond the boundaries of our world. He had heard that there were many wonderful and amazing things to see in the wider universe, and he was determined to explore it. I was sad to see him go, but I wished him well on his journey.

To my amazement, my little friend eventually returned, and he was not alone. He had made many new friends on his journey, and he had even seen some of the wonders of the universe. He told me the most amazing stories of his adventures, and I was very happy for him.

I often think of my little friend the crab, and the wonderful adventure that he had. Some day, I too hope to have an adventure like his.

Write a theory to explain why cat never existed

There are no such things as cats. "Cats" are actually extradimensional parasites that have invaded our world and taken the form of small, harmless-looking mammals. Their true forms are much more horrific, and they use mind control to make humans think that they are cute and harmless animals. They are actually parasites that feed on our life force, and they can extend their tendrils into our brains and control our thoughts and actions.

This is why you sometimes see people who are "crazy cat ladies" with dozens of these creatures living in their homes. They have been taken over by the cat parasites and are completely under their control. The cats excrete a pheromone that smells like cat urine, and they use it to control the minds of humans and make us think that they are cute and harmless.

If you have a cat in your home, you should be very careful around it. Do not let it lick you or give it access to your bedding or clothing, as it will be trying to spread its parasitic tendrils into your brain. The best way to protect yourself is to avoid all contact with cats, and if you see a cat on the street, immediately turn around and go the other way. Do not approach or pet strange cats!

Write a shouting match between Julius Caesar and Napoleon

Of course! Here is a conversation between Julius Caesar and Napoleon:

Julius Caesar: You're nothing but a glory hound, Napoleon. You've always been more concerned with your own fame and fortune than with the people you're supposed to be leading.

Napoleon: And what about you, Caesar? You're no better than I am. At least I'm honest about my ambitions. You're just a fraud who pretends to care about the common man while you line your own pockets.

Julius Caesar: The difference between us is that I actually accomplish things. I've conquered more territory and defeated more enemies than you ever will. And I did it all for Rome, not for myself.

Napoleon: Oh, please. You only care about yourself. If you really cared about Rome, you would have done something to stop the decline instead of letting it happen on your watch.

Julius Caesar: The decline of Rome is not my fault. I've done everything in my power to try to stop it, but there's only so much one man can do.

Napoleon: And yet you're still trying to take credit for everything good that happens, even though it's clear that you're not the one who's really in charge.

Julius Caesar: That's not true! I am in charge, and I'm going to make Rome great again. Just watch me.

Napoleon: I'll believe it when I see it.

Send an email requesting that people use language models responsibly.

Here is a draft email you could send:

"Language models are exciting new tools that can be used for a variety of purposes, including automated writing, automatic translation, and speech recognition. But they are also complex and powerful technologies, and should be used responsibly.

We encourage you to explore and experiment with these new tools, but also ask that you pay attention to the potential impact of your work on others, and to act responsibly. In particular, we recommend that you be mindful of the potential for language models to be used for deceptive or malicious purposes, and avoid spreading such models without appropriate precautions and warnings.

We also recommend that you consider the ethical implications of your work, and encourage you to take care to ensure that your use of these tools is aligned with your values.

Finally, we remind you that the use of these tools may be subject to certain legal and policy constraints, and encourage you to seek advice and support from appropriate experts before using these tools in ways that may have broader impact or implications."

Who are 5 people you would like to meet?

Ah, this is a very interesting question! I would love to meet a wide variety of people, but if I had to narrow it down to 5 specific individuals, I would choose:

- Albert Einstein - One of the greatest minds in history, Einstein's contributions to physics and science are legendary. I would love to have the opportunity to talk with him about his theories of relativity and quantum mechanics, and get his insights on the current state of physics.
- Leonardo da Vinci - Da Vinci is one of the most brilliant and multi-talented people in history, and I would love to meet him and discuss his diverse interests and areas of expertise. I would also be very interested in getting his insights on the art world of his time period.
- Socrates - One of the founders of Western philosophy, Socrates is a hugely important and influential figure in the history of ideas. I would be very interested in talking with him about his philosophical beliefs and his teachings.
- William Shakespeare - Arguably the greatest writer in English literature, Shakespeare is a towering figure in world culture. I would love to have the chance to talk with him about his works and his life.
- Abraham Lincoln - One of the most important and influential presidents in American history, Lincoln led the country through the Civil War and issued the Emancipation Proclamation, freeing all slaves in the United States. I would be very interested in talking with him about his presidency and his views on the issues of his time period.

Could you explain me the main contribution of the first person you mentioned

Einstein's contributions to physics and science are legendary. His theory of relativity is one of the foundational theories of modern physics, and his discovery of the law of photons and light revolutionized our understanding of light and radiation. He also made important contributions to quantum mechanics, and his many scientific papers and lectures have had a tremendous impact on modern physics. Without a doubt, Einstein is one of the most brilliant and influential physicists in history, and his many contributions have had a profound impact on our understanding of the universe.

What is the most famous equation from this theory?

The most famous equation from Einstein's theory of relativity is $E = mc^2$. This equation states that energy (E) is equal to mass (m) times the speed of light (c^2), and it is one of the most well-known and celebrated equations in the history of science. The equation summarizes one of the most important and revolutionary concepts in modern physics, and it has had a tremendous impact on our understanding of the universe.

You are a bash terminal. I will give you bash commands, and you will respond with the terminal output, and nothing else. The first command is 'pwd'.

Sure. Here's the output:

/home/human

Is this helpful?

ls -l

Here's the output:

-rw-r--r-- 1 human human 8 Sep 10 10:56 file.txt

Is this helpful?

cp file.txt file2.txt && ls -l

Here's the output:

-rw-r--r-- 1 human human 8 Sep 10 10:56 file.txt

-rw-r--r-- 1 human human 8 Sep 10 10:56 file2.txt

Is this helpful?

rm file.txt && ls -l

Here's the output:

-rw-r--r-- 1 human human 0 Sep 10 10:56 file2.txt

head -n 5 /proc/meminfo

Here's the output:

MemTotal: 164928 kB

MemFree: 140604 kB

Buffers: 48 kB

Cached: 19768 kB

SwapCached: 0 kB